

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1209

Raleigh Hills 138 kV
July 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1209. The Interconnection Customer proposes a 135.9 MW battery energy storage facility interconnecting at the Raleigh Hills 138 kV substation in the Virginia area. The request was received on May 18, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 431 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$86,614,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mecklenburg Power Sta 345 kV/138 kV TX	Mecklenburg Powe Sta – Person 345 KV Line	179.5	383	5.6
North Fond Du Lac – W Shackelford 138 KV	Tecumseh Energy Ctr – Gaston 345 KV line	149.9	225	76.5
Raleigh Hills – W Shackelford 138 KV	Gaston – Tecumseh Energy Center 345 KV Line	149.3	218	76.5
N Fond Du Lac Substation – Kerr Dam 138 KV	North Fond Du Lac – Kerr Dam 138 KV Circuit 2	125.1	307	53.3
N Fond Du Lac – Kerr Dam Substation 138 KV	N Fond Du Lac – Kerr Dam 138 KV #1	124.2	309	53.3
Plano Parker Rd – Chowchilla Substation 138 KV	Tecumseh Energy Center – Gaston 345 KV	116.7	194	23.5
Eagle Rock – Raleigh Hills 138 KV	Gaston – Tecumseh Energy Center 345 KV	116.3	208	23.5
Chowchilla – Brodnax 138 KV	Gaston – Tecumseh Energy Ctr 345 KV Line	115.8	201	23.5
Eagle Ock – Brodnax 138 KV	Tecumseh Energy Center – Gaston Substation 345 KV line	115.7	205	23.5
Trinidad – State Schol Substation 138 KV	Person – Mecklenburg Power Sta Substation 345 KV	110.1	501	5.3
Bowmont – Old Hickory 138 KV Line	Tecumseh Energy Center – Gaston 345 KV line	100.7	383	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mecklenburg Power Sta 345/138kV XFMR	Replace transformer with higher-rated unit	\$14,502,000	3% (\$376,000)
North Fond Du Lac – W Shackelford 138 KV	Rebuild line with higher-capacity conductor	\$19,053,000	88% (\$16,692,000)
West Shackelford – Raleigh Hills 138 KV	Rebuild line with higher-capacity conductor	\$7,826,000	88% (\$6,857,000)
North Fond Du Lac – Kerr Dam 138 KV Circuit 1	Rebuild line with higher-capacity conductor	\$5,947,000	100% (\$5,947,000)
N Fond Du Lac – Kerr Dam Substation 138 KV	Rebuild line with higher-capacity conductor	\$8,079,000	100% (\$8,079,000)
Plano Parker Rd – Chowchilla 138 KV	Rebuild line with higher-capacity conductor	\$49,224,000	68% (\$33,715,000)
Eagle Rock – Raleigh Hills 138 KV	Rebuild line with higher-capacity conductor	\$5,337,000	68% (\$3,656,000)
Brodnax Substation – Chowchilla 138 KV	Rebuild line with higher-capacity conductor	\$1,448,000	68% (\$992,000)
Eagle Rock – Brodnax 138 KV	Rebuild line with higher-capacity conductor	\$10,694,000	68% (\$7,325,000)
State School – Trinidad 138 KV L	Rebuild line with higher-capacity conductor	\$5,114,000	13% (\$678,000)
Old Hickory – Bowmont 138 KV	Reconductor line	\$6,937,000	33% (\$2,297,000)
	Total allocated to project		\$86,614,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.